



Embedded Software for the Internet Of Things

Introduction to TI MSP432
Launchpad

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Embedded System Development Platform

- **Host system**

- cross compiler, linker, and source-level debugger.

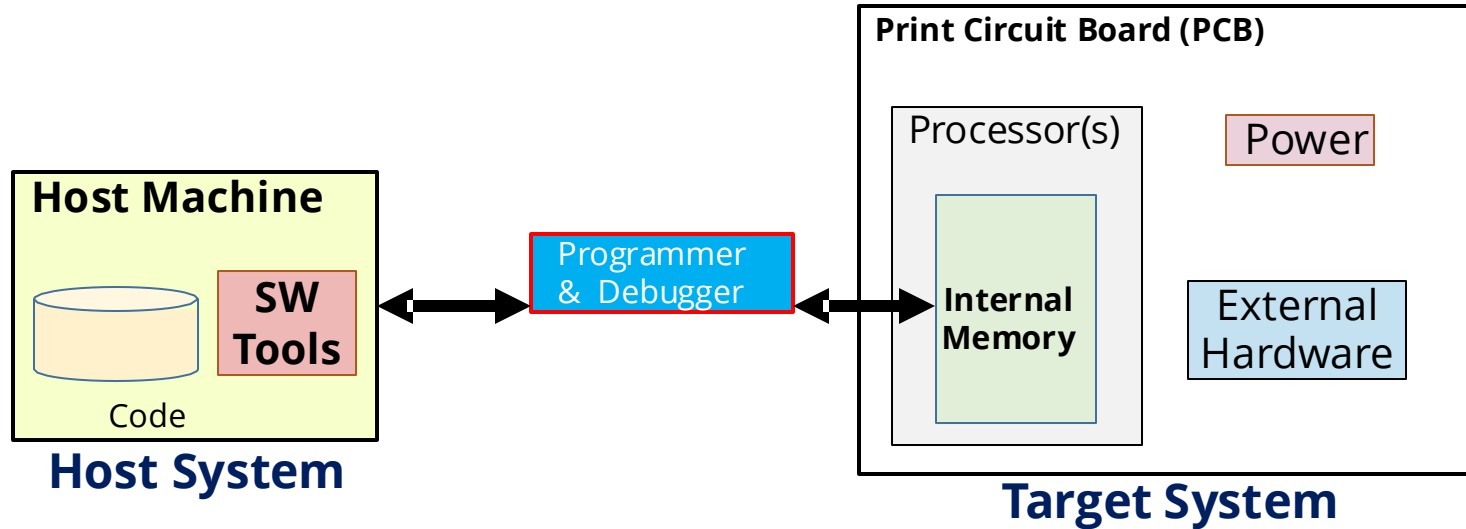
- **The target**

- debugger, processor, sensors,...

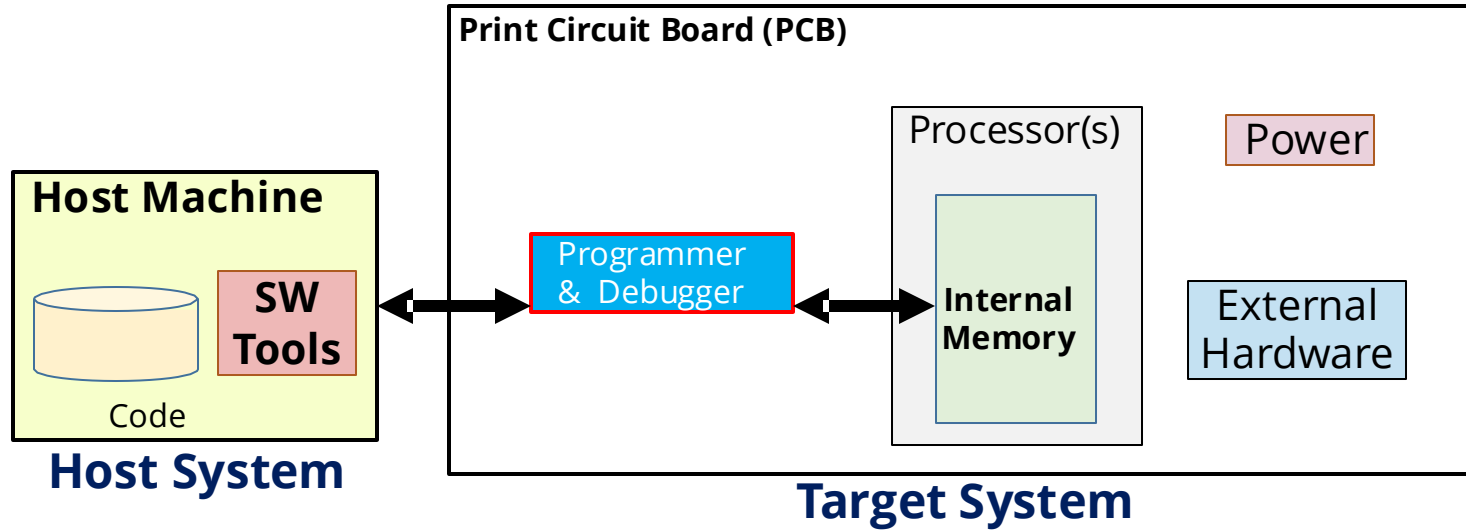
- **Connections** between the host and the target system:

- For downloading program images from the host system to the target system.
- For transmitting **debugger information** between the host debugger and the target debug agent.

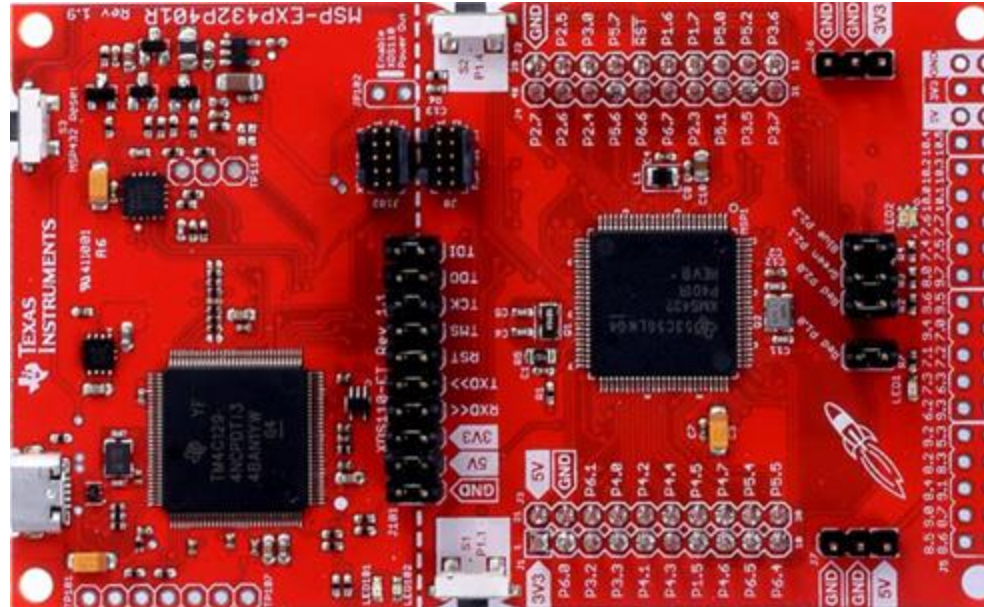
Embedded System Development Platform



Embedded System Development Platform

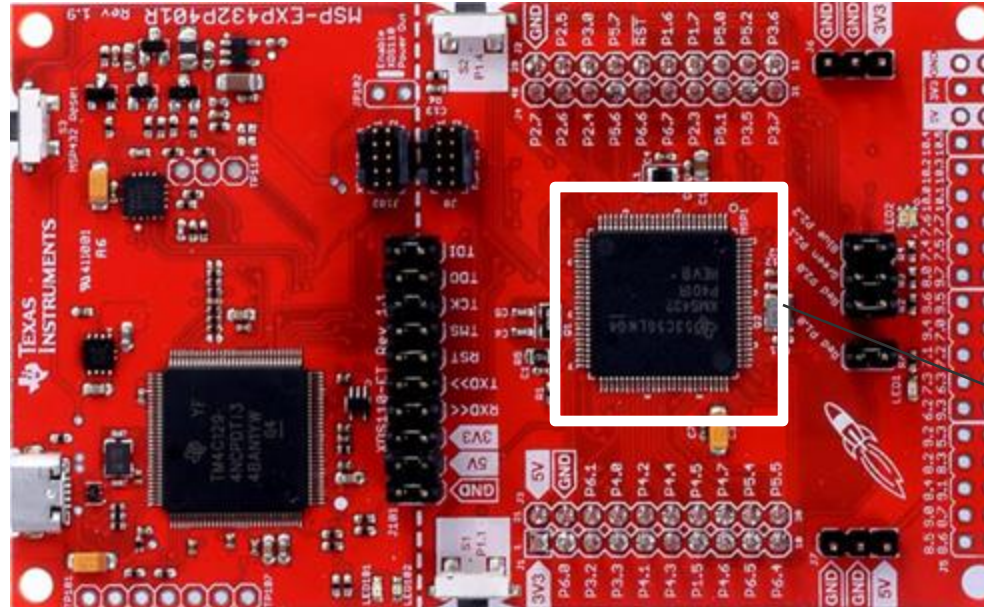


Embedded System Development Platform



Printed
Circuit
Board

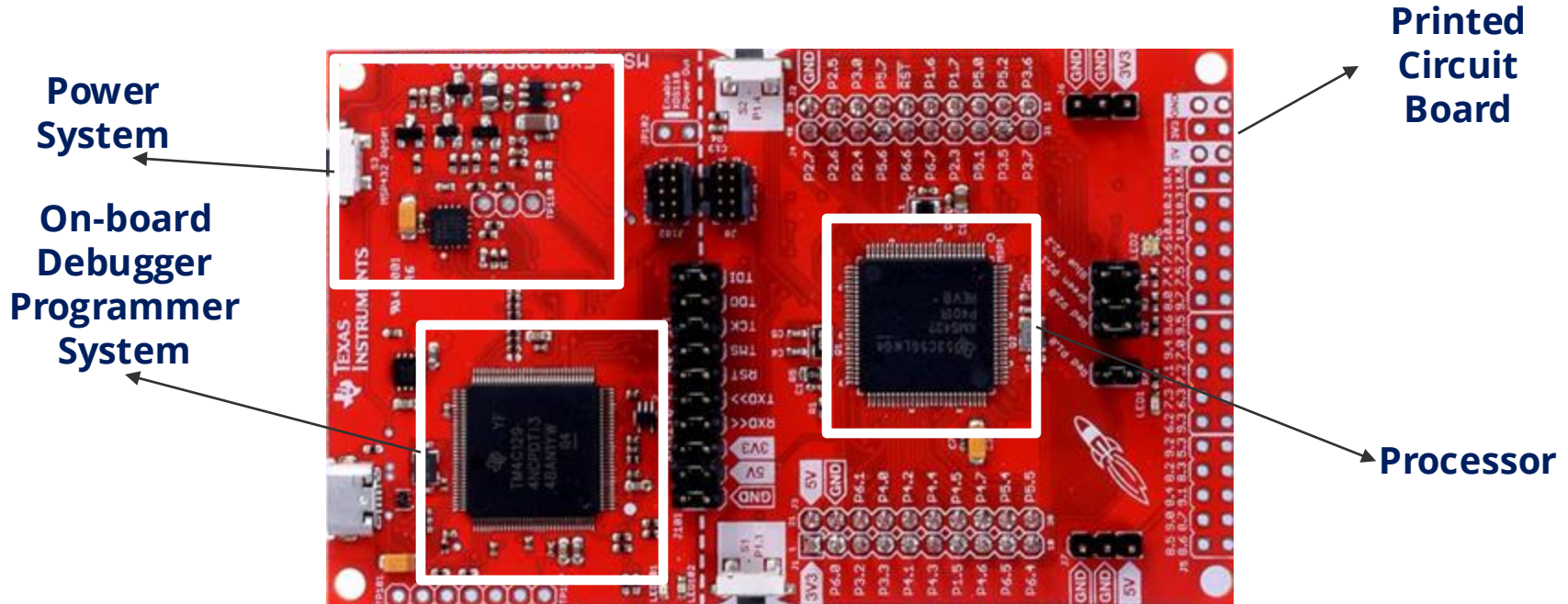
Embedded System Development Platform



**Printed
Circuit
Board**

Processor

Embedded System Development Platform



Embedded System Development Platform

Power System

On-board
Debugger
Programmer
System

Host Machine

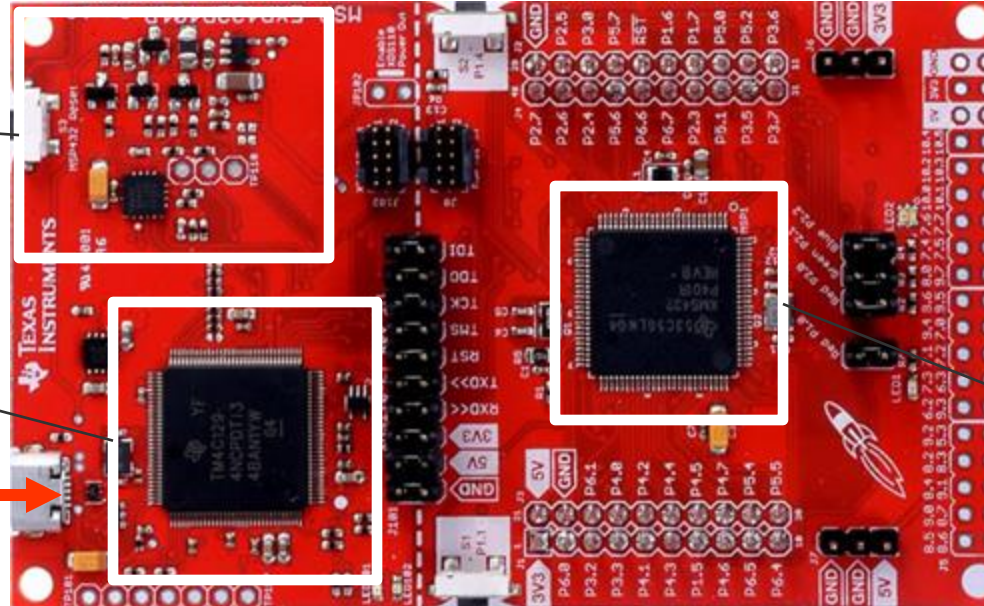
SW
Tools

Code

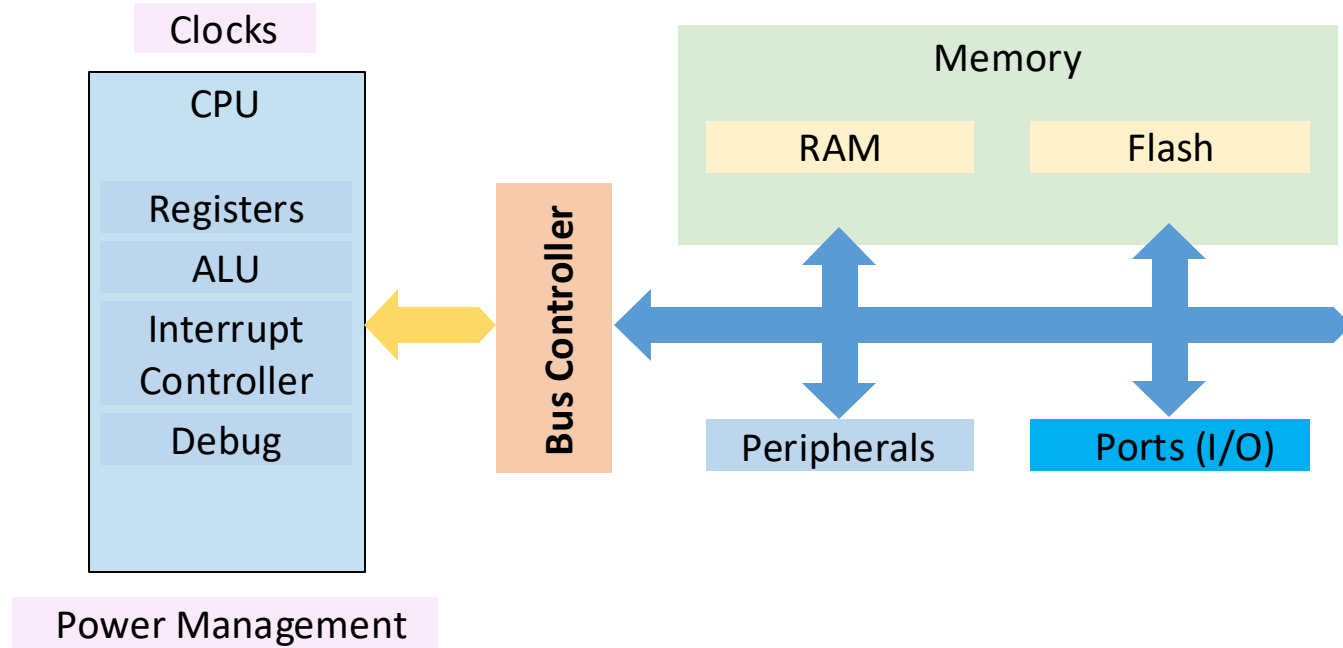
Host System

Printed
Circuit
Board

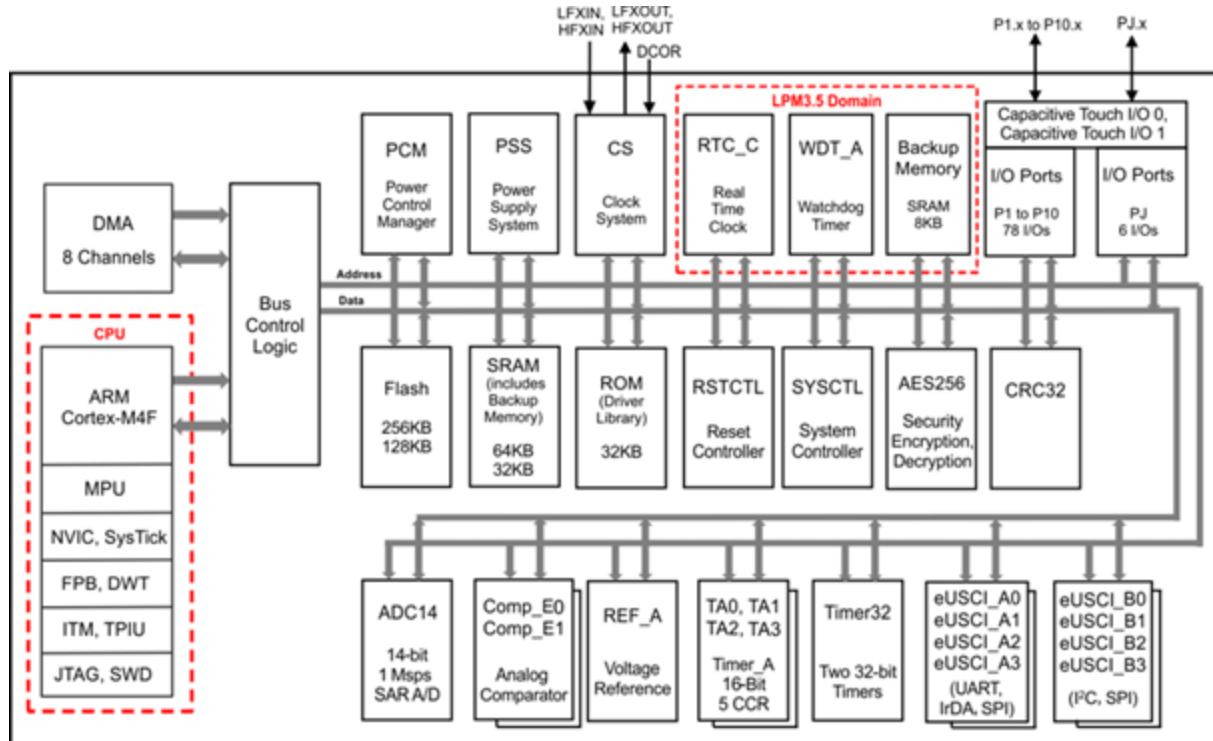
Processor



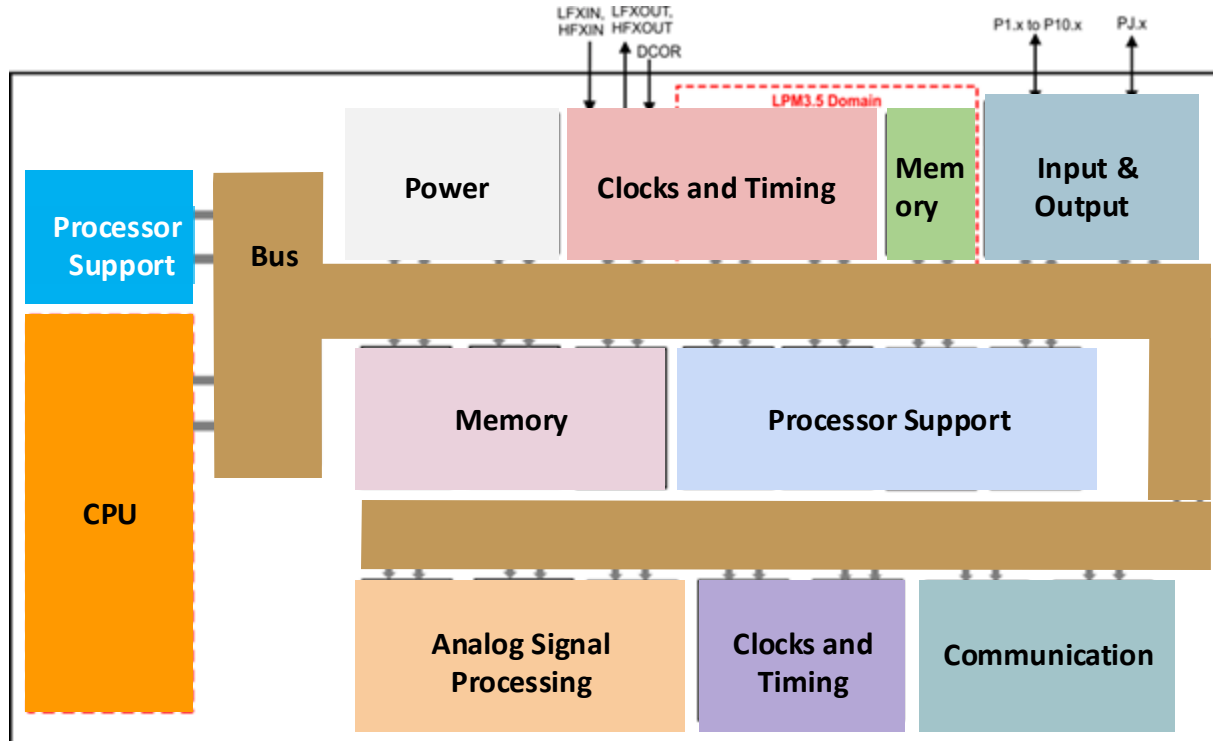
Microcontroller Components



MSP432 Cortex-M4 Microcontroller



MSP432 Cortex-M4 Microcontroller



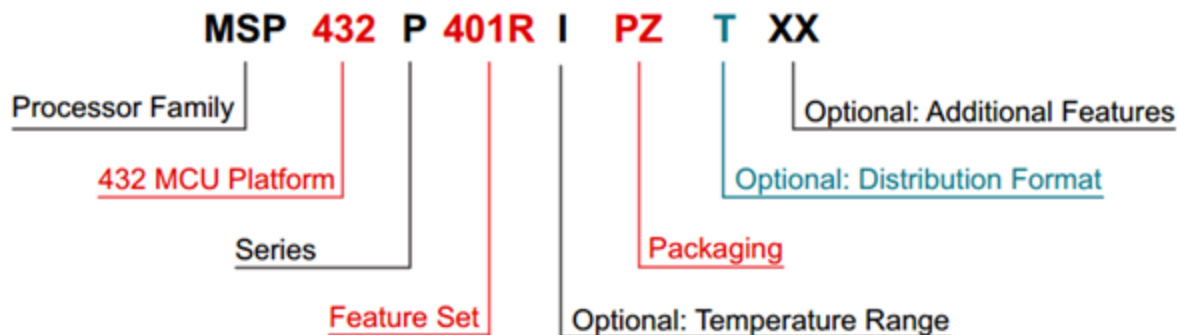
MSP432P401R Main Features

- ARM® Cortex™-M4F processor
- 100-pin microcontroller chip
- MSP stands for Mixed Signal Processor (analog plus digital)
- 256K bytes (256KB) of on-chip Flash memory for code
- 64KB of on-chip SRAM for data
- A large number of on-chip peripherals

Memory Map in MSP432401P401R

Allocated size		Allocated address
Flash	256KB	0x0000 0000 to 0x0003 FFFF
SRAM	64KB	0x2000 0000 to 0x2000 FFFF
I/O	All the peripherals	0x4000 0000 to 0x4001 FFFF

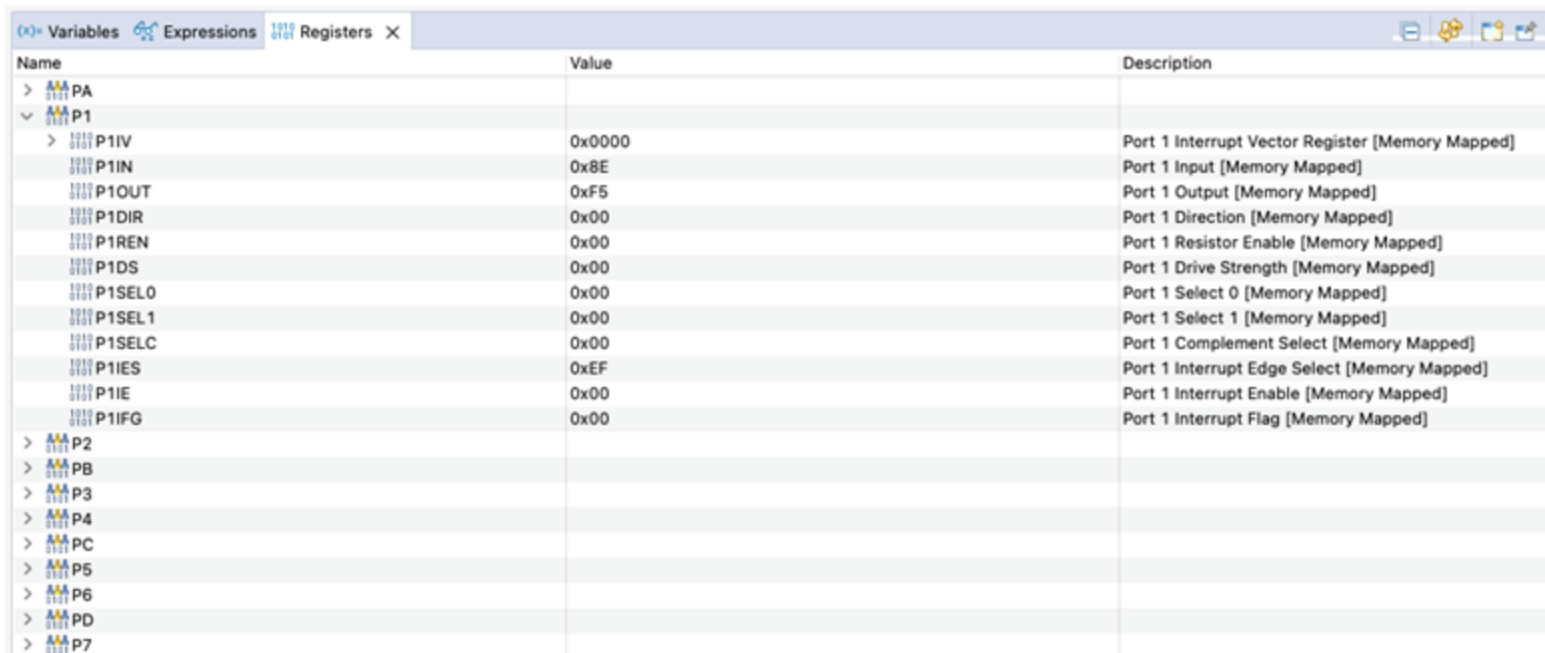
TI'S Naming Convention For MSP432



Processor Family	MSP = Mixed Signal Processor XMS = Experimental Silicon			
432 MCU Platform	TI's 32-bit Low-Power Microcontroller Platform			
Series	P = Performance and Low-Power Series			
Feature Set	First Digit 4 = Flash based devices up to 48 MHz	Second Digit 0 = General Purpose	Third Digit 1 = ADC14	Fourth Digit R = 256KB of Flash. 64KB of SRAM M = 128KB of Flash, 32KB of SRAM
Optional: Temperature Range	S = 0°C to 50 °C I = 40 °C to 85 °C T = -40 °C to 105 °C			
Packaging	PZ = LQFP			

TI Code Composer Studio (CCS)

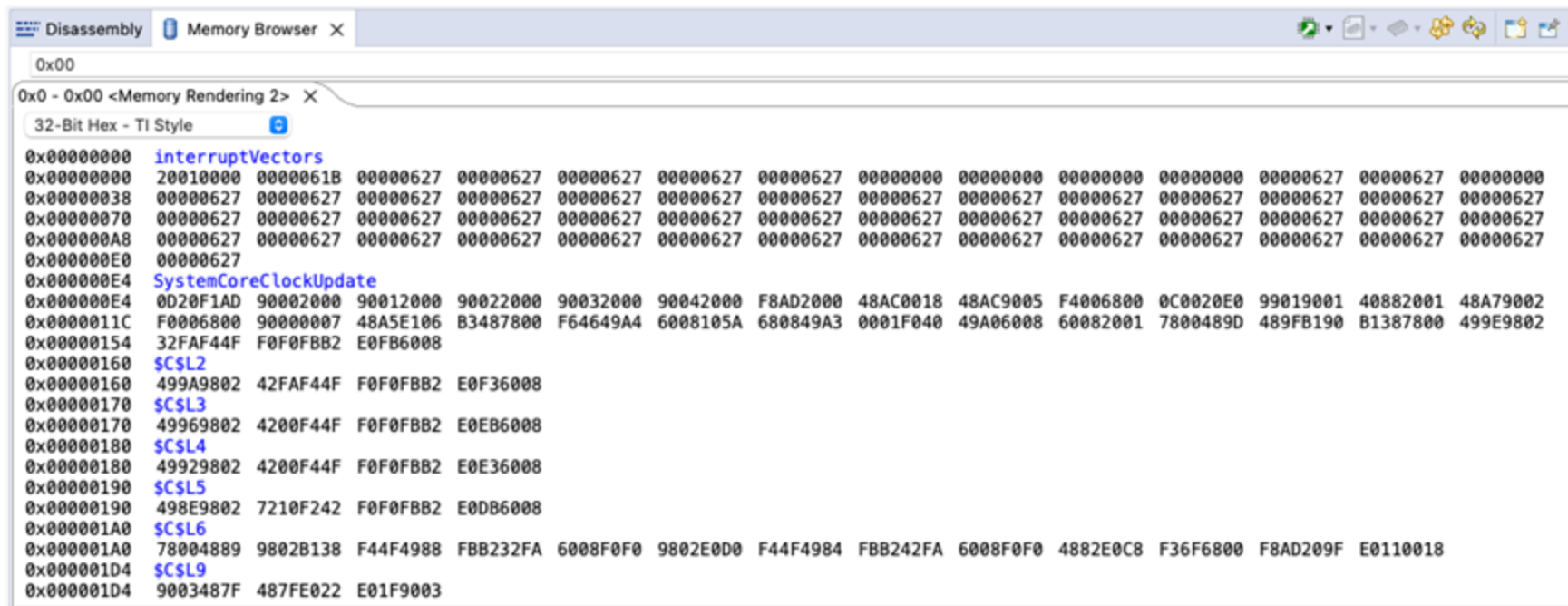
- Variables, Expressions, Registers



Name	Value	Description
> PA		
▼ P1		
> P1IV	0x0000	Port 1 Interrupt Vector Register [Memory Mapped]
P1IN	0x8E	Port 1 Input [Memory Mapped]
P1OUT	0xF5	Port 1 Output [Memory Mapped]
P1DIR	0x00	Port 1 Direction [Memory Mapped]
P1REN	0x00	Port 1 Resistor Enable [Memory Mapped]
P1DS	0x00	Port 1 Drive Strength [Memory Mapped]
P1SEL0	0x00	Port 1 Select 0 [Memory Mapped]
P1SEL1	0x00	Port 1 Select 1 [Memory Mapped]
P1SELC	0x00	Port 1 Complement Select [Memory Mapped]
P1IES	0xEF	Port 1 Interrupt Edge Select [Memory Mapped]
P1IE	0x00	Port 1 Interrupt Enable [Memory Mapped]
P1IFG	0x00	Port 1 Interrupt Flag [Memory Mapped]
> P2		
> PB		
> P3		
> P4		
> PC		
> P5		
> P6		
> PD		
> P7		

TI CCS

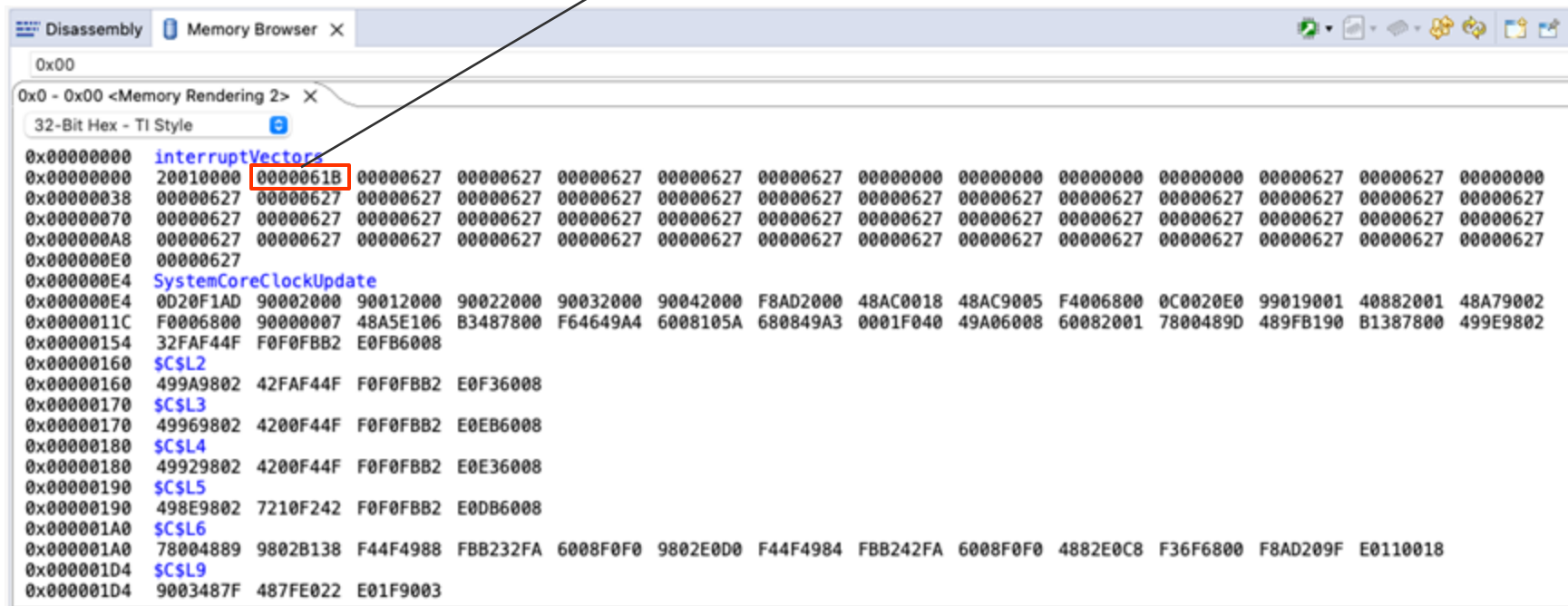
- Interrupt Vectors (0x0)



TI CCS

- Interrupt Vectors (0x0)

Reset Vector



TI CCS

- Reset Vector (0x0000061B)

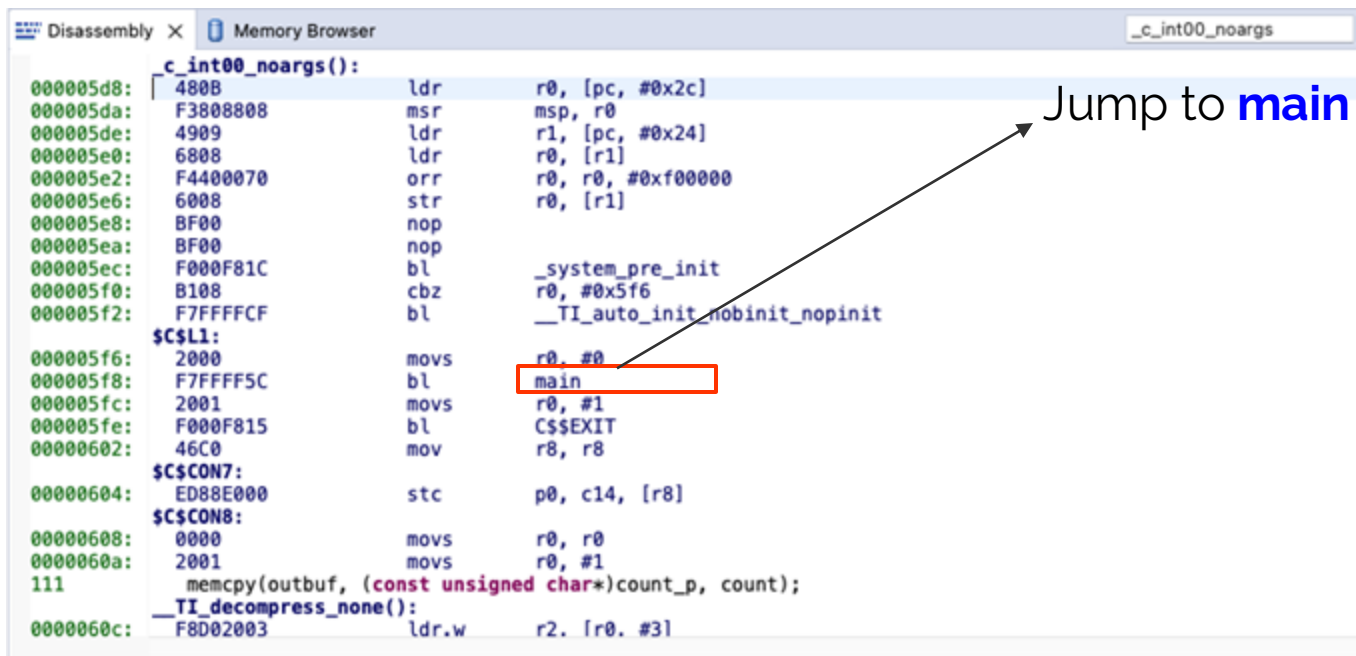
Jump to
[_c_int00_noargs](#)

```
Disassembly X Memory Browser 0x0000061B

Reset_Handler():
0000061a: B508      push    {r3, r14}
183      SystemInit();
0000061c: F7FFFE9E  bl       SystemInit
186      _asm("    .global _c_int00\n"
00000620: F7FFBFDA  b.w      _c_int00_noargs
188      }
00000624: BD08      pop     {r3, pc}
201      while(1)
$C$L1, Default_Handler():
00000626: E7FE      b        Default_Handler
58      return 1;
_system_pre_init():
00000628: 2001      movs    r0, #1
0000062a: 4770      bx      r14
105      {
C$$EXIT(), abort():
0000062c: BF00      nop
124      for (;;) /* SPINS FOREVER */
$C$L1:
0000062e: E7FE      b        $C$L1
00000630: 0F00      lsrs    r0, r0, #0x1c
00000632: C6C0      stm     r6!, {r6, r7}
00000634: 002D      movs    r5, r5
00000636: F0FF052D .word    0x052df0ff
00000638: 0000      movs    r4, r4
```

TI CCS

- `_c_int00_noargs`



The screenshot shows the Disassembly window in TI CCS. The function `_c_int00_noargs()` is displayed, starting at address `000005d8`. The assembly code includes instructions for loading the program counter, setting the main stack pointer, and performing various conditional branches. A red box highlights the `main` label at address `000005f8`, which is the target of a `bl` (branch link) instruction. An arrow points from this label to the text "Jump to **main**".

```
Disassembly X Memory Browser _c_int00_noargs

_c_int00_noargs():
000005d8: 480B      ldr      r0, [pc, #0x2c]
000005da: F3808808  msr      msp, r0
000005de: 4909      ldr      r1, [pc, #0x24]
000005e0: 6808      ldr      r0, [r1]
000005e2: F440070  orr      r0, r0, #0xf00000
000005e6: 6008      str      r0, [r1]
000005e8: BF00      nop
000005ea: BF00      nop
000005ec: F00F81C  bl       _system_pre_init
000005f0: B108      cbz      r0, #0x5f6
000005f2: F7FFFCF  bl       __TI_auto_init_nobinit_nopinit

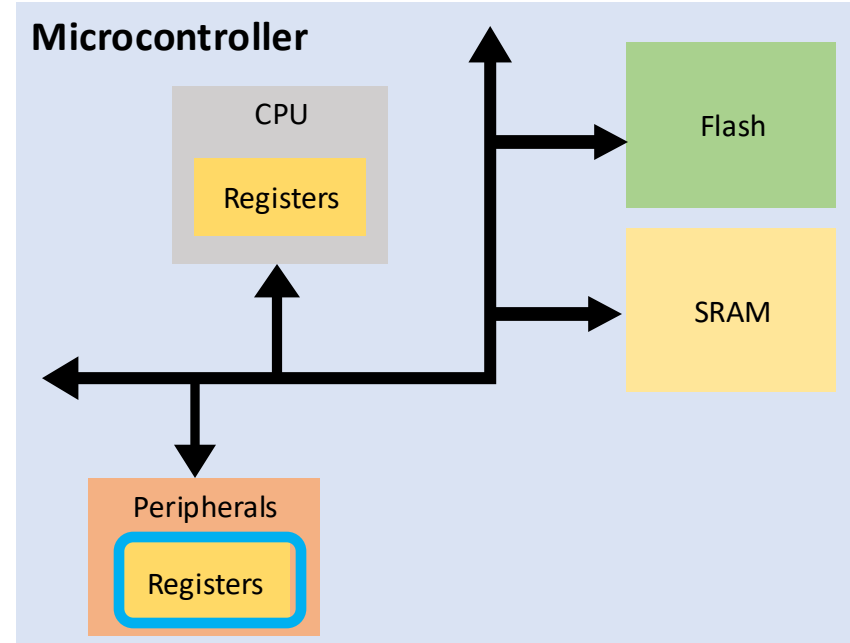
$CS$L1:
000005f6: 2000      movs     r0, #0
000005f8: F7FFF5C  bl       main
000005fc: 2001      movs     r0, #1
000005fe: F00F815  bl       CS$EXIT
00000602: 46C0      mov      r8, r8

$CS$CON7:
00000604: ED88E000 stc      p0, c14, [r8]

$CS$CON8:
00000608: 0000      movs     r0, r0
0000060a: 2001      movs     r0, #1
111      memcpy(outbuf, (const unsigned char*)count_p, count);
__TI_decompress_none():
0000060c: F8D02003 ldr.w    r2, [r0, #3]
```

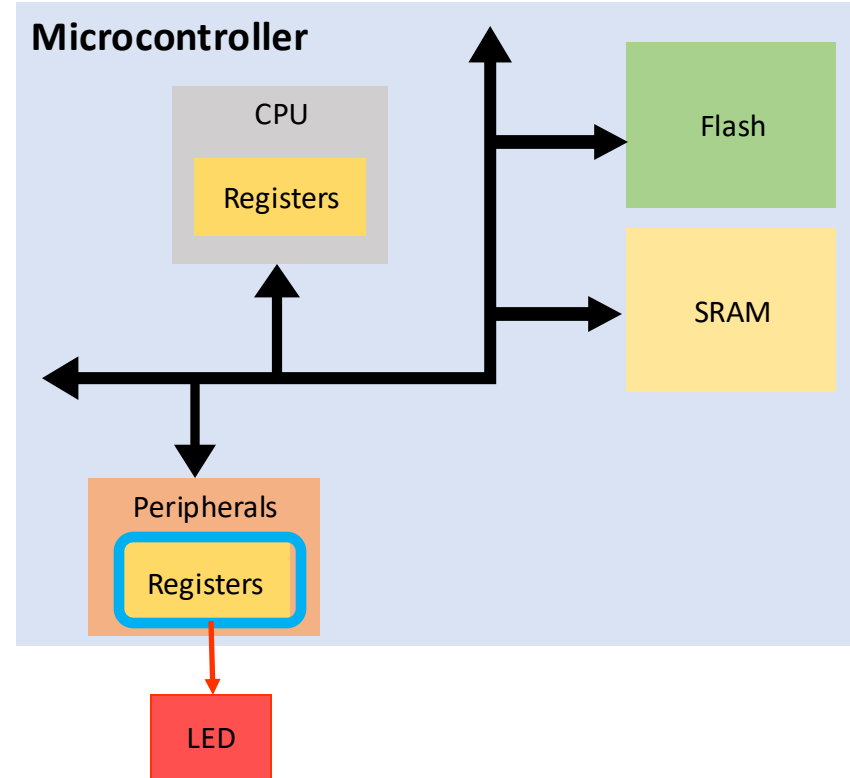
Peripherals

- Peripherals: External to CPU
 - Memory-mapped I/O to configure peripherals
 - Pointers
 - Memory Reads/Writes
 - Use of Bit Manipulation



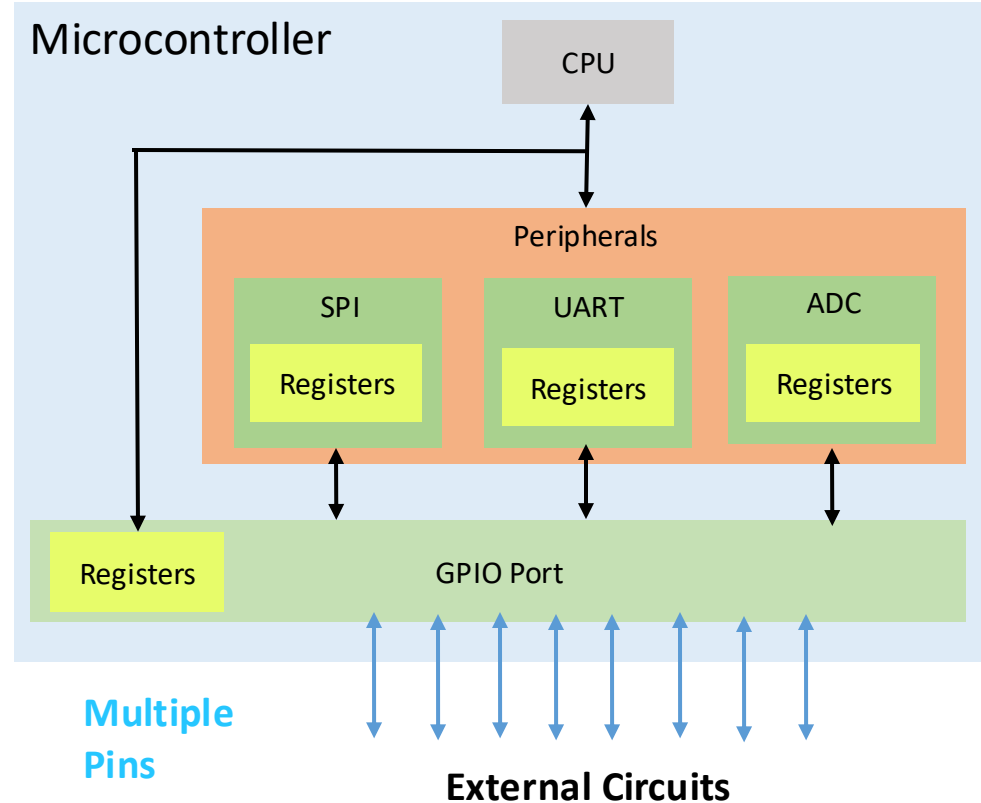
Peripherals

- Peripherals: External to CPU
 - Memory-mapped I/O to configure peripherals
 - Pointers
 - Memory Reads/Writes
 - Use of Bit Manipulation
- GPIO - General Purpose IO



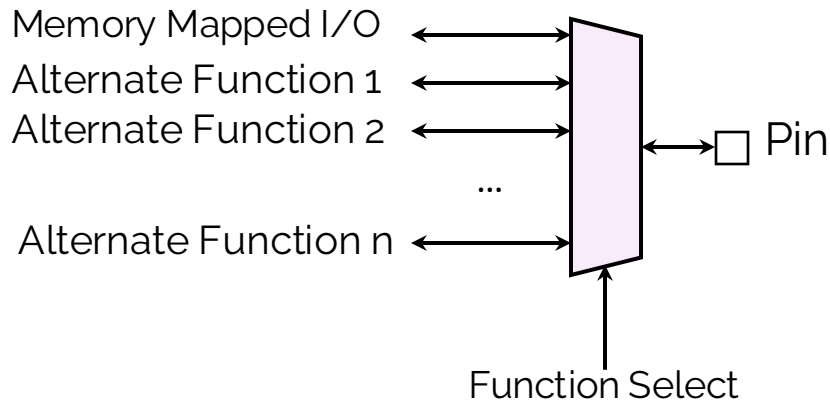
Input/Output Systems

- Input/Output:
 - method to get data in/out of microcontroller
- GPIO-General Purpose IO
 - Pin – physical connection to microcontroller
 - Port – combination of pins

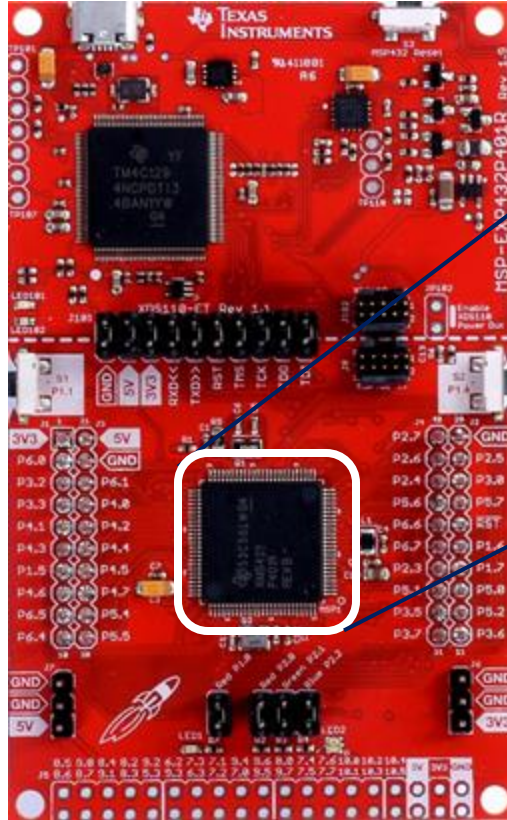


GPIO Alternative Functions

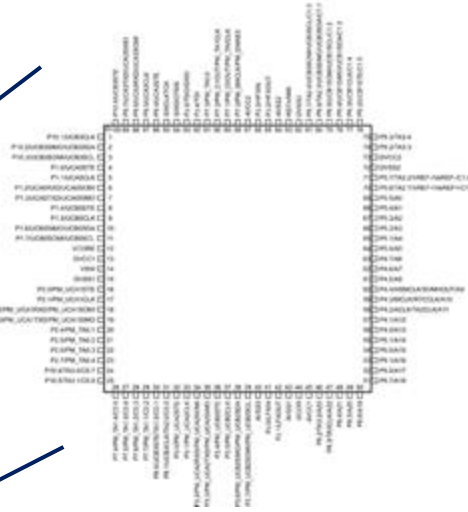
- Pins may have different features
- To enable an alternative function, set up the appropriate register



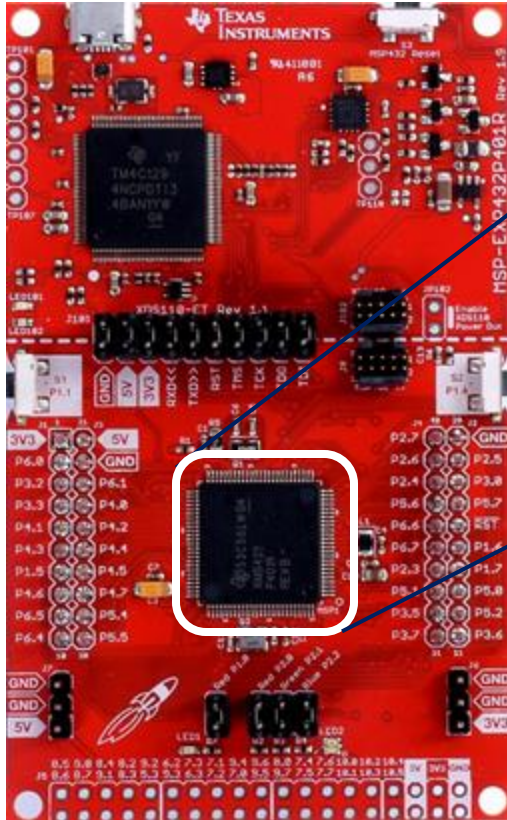
MSP432 Ports



10 Ports



MSP432 Ports



10 Ports

Port 1 Pin 0 = P1.0

Primary Mode = SPI Chip Select

Port 1 Pin 7 = P1.7

Primary Mode = SPI Slave-Out-Master-In Data (**UCB0SOMI**)

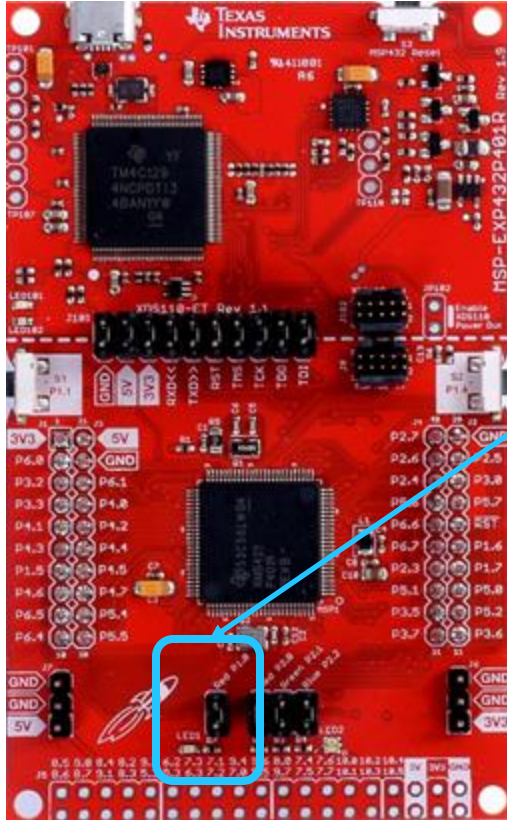
Secondary Mode = I²C Clock (**UCB0SCL**)

P1.0/UCAB0STE	☐	4
P1.1/UCAB0CLK	☐	5
P1.2/UCAB0RXD/UCAB0SOMI	☐	6
P1.3/UCAB0TXD/UCAB0SIMO	☐	7
P1.4/UCB0STE	☐	8
P1.5/UCB0CLK	☐	9
P1.6/UCB0SIMO/UCB0SDA	☐	10
P1.7/UCB0SOMI/UCB0SCL	☐	11

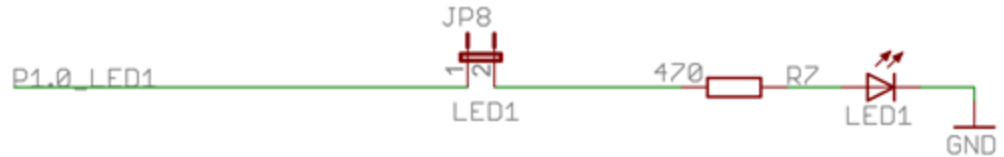
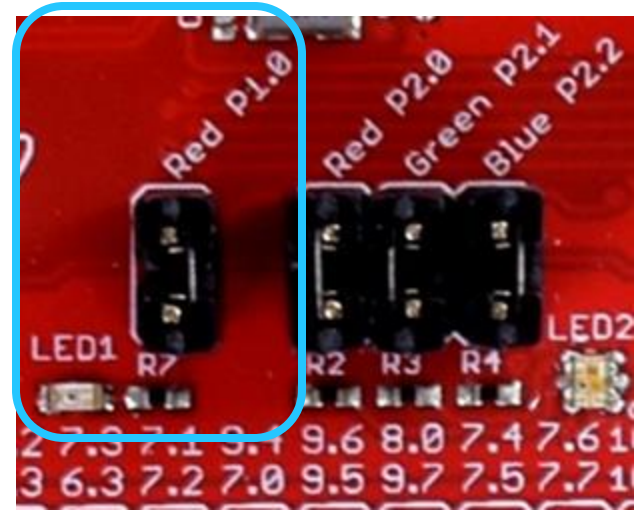
Port 1 – 8 Pins

GPIO / Primary / Secondary

MSP432 Ports

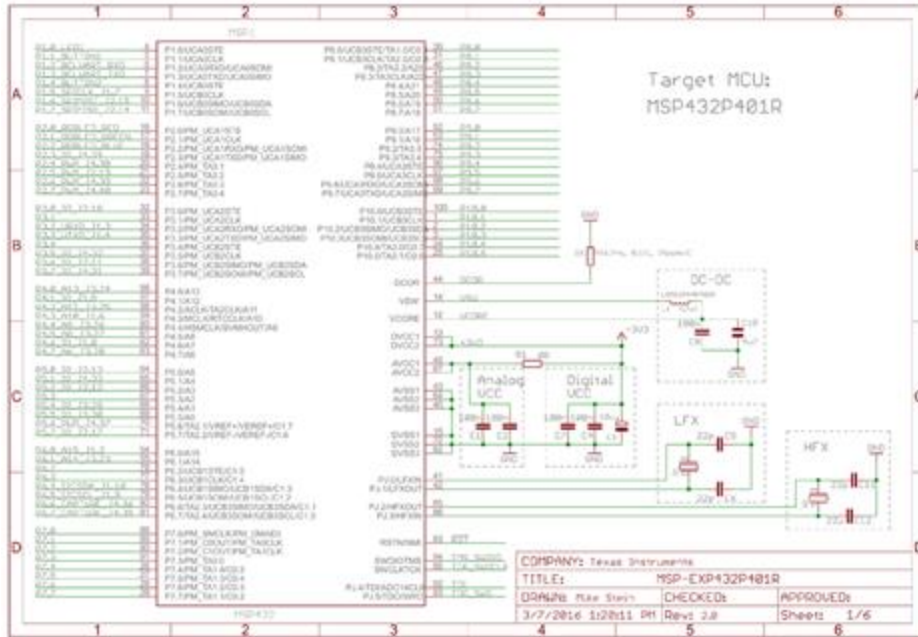


P1.0 Red LED Circuit



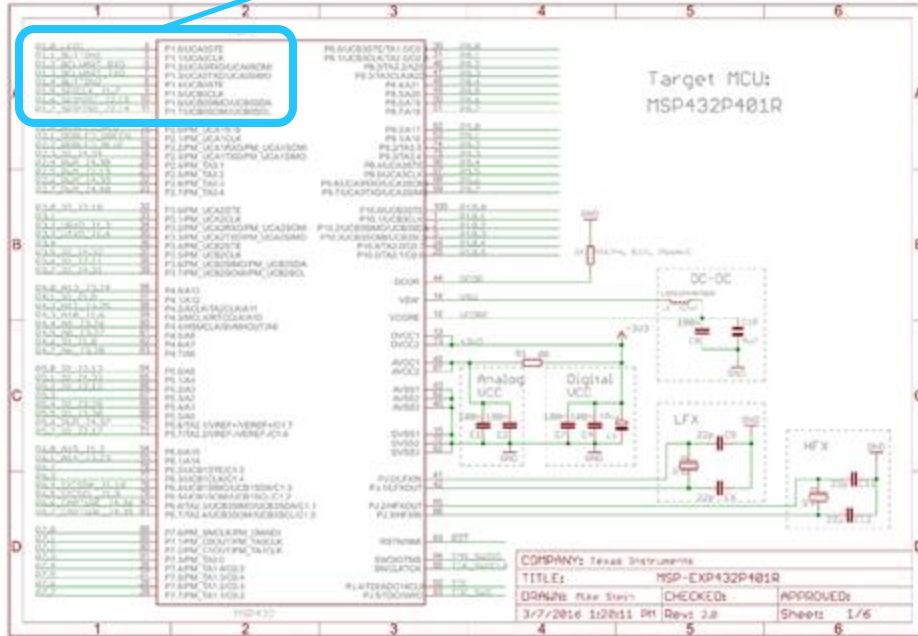
LED Schematic for P1.0

MSP432 Launchpad Schematic



MSP432 Launchpad Schematic

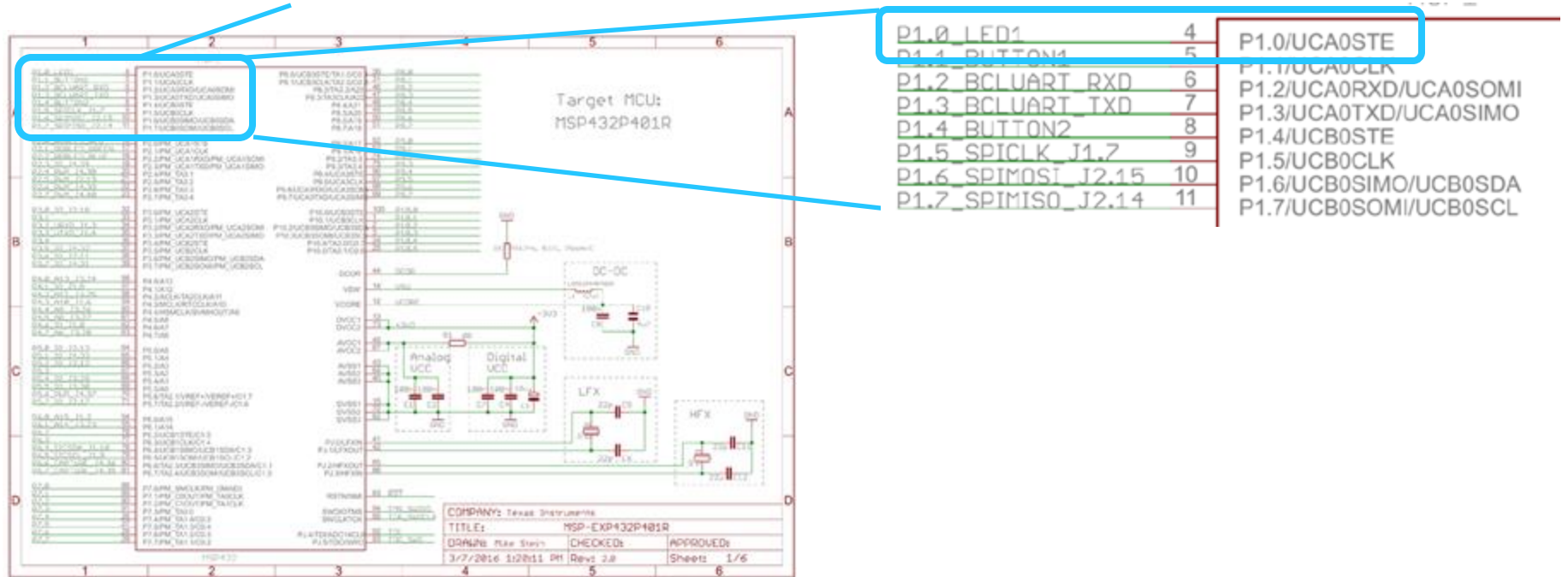
MSP432 Port 1



Launchpad Schematic: MSP432 Pinout

MSP432 Launchpad Schematic

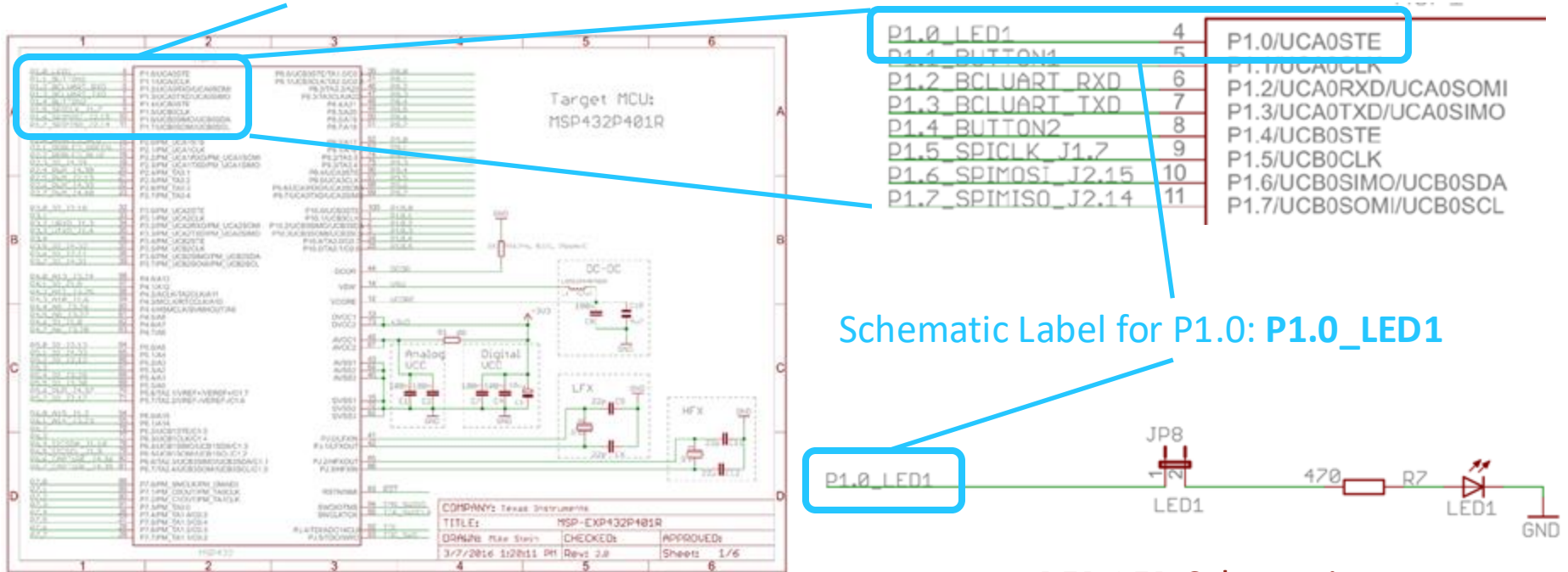
MSP432 Port 1



Launchpad Schematic: MSP432 Pinout

MSP432 Launchpad Schematic

MSP432 Port 1



Launchpad Schematic: MSP432 Pinout

RED LED Schematic

MSP432 Launchpad Schematic

MSP432 Pinout Port 1

P1.0_LED1	4	P1.0/UCAS0STE
P1.1_BUTTON1	5	P1.1/UCAS0CLK
P1.2_BCUART_RXD	6	P1.2/UCAS0RXD/UCAS0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAS0TXD/UCAS0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIMOSI_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

MSP432 Pin Descriptions

Table 6-32. Port P1 (P1.0 to P1.7) Pin Functions

PIN NAME (P1.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P1DIR.x	P1SEL1.x	P1SEL0.x
P1.0/UCAS0STE	0	P1.0 (I/O)	I: 0; O: 1	0	0
		UCAS0STE	X ⁽²⁾	0	1
		N/A	0	1	0
		DVSS	1		
		N/A	0	1	1
		DVSS	1		
P1.1/UCAS0CLK	1	P1.1 (I/O)	I: 0; O: 1	0	0
		UCAS0CLK	X ⁽²⁾	0	1
		N/A	0	1	0
		DVSS	1		
		N/A	0	1	1
		DVSS	1		
P1.2/UCAS0RXD/UCAS0SOMI	2	P1.2 (I/O)	I: 0; O: 1	0	0
		UCAS0RXD/UCAS0SOMI	X ⁽²⁾	0	1
		N/A	0	1	0
		DVSS	1		
		N/A	0	1	1
		DVSS	1		

MSP432 Launchpad Schematic

MSP432 Pinout Port 1

P1.0_LED1	4	P1.0/UCAS0STE
P1.1_BUTTON1	5	P1.1/UCAS0CLK
P1.2_BCUART_RXD	6	P1.2/UCAS0RXD/UCAS0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAS0TXD/UCAS0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIMOSI_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Pin
Number

MSP432 Pin Descriptions

Table 6-32. Port P1 (P1.0 to P1.7) Pin Functions

PIN NAME (P1.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P1DIR.x	P1SEL1.x	P1SEL0.x
P1.0/UCAS0STE	0	P1.0 (I/O)	I: 0; O: 1	0	0
		UCAS0STE	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.1/UCAS0CLK	1	P1.1 (I/O)	I: 0; O: 1	0	0
		UCAS0CLK	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.2/UCAS0RXD/UCAS0SOMI	2	P1.2 (I/O)	I: 0; O: 1	0	0
		UCAS0RXD/UCAS0SOMI	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1

Port P1Sel0.x / P1Sel1.x registers
Select Mode for each Pin in Port

MSP432 Launchpad Schematic

MSP432 Pinout Port 1

P1.0_LED1	4	P1.0/UCAS0STE
P1.1_BUTTON1	5	P1.1/UCAS0CLK
P1.2_BCUART_RXD	6	P1.2/UCAS0RXD/UCAS0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAS0TXD/UCAS0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIMOSI_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Pin
Number

Pin
Modes

MSP432 Pin Descriptions

Table 6-32. Port P1 (P1.0 to P1.7) Pin Functions

PIN NAME (P1.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P1DIR.x	P1SEL1.x	P1SEL0.x
P1.0/UCAS0STE	0	P1.0 (I/O)	I: 0; O: 1	0	0
		UCAS0STE	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.1/UCAS0CLK	1	P1.1 (I/O)	I: 0; O: 1	0	0
		UCAS0CLK	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.2/UCAS0RXD/UCAS0SOMI	2	P1.2 (I/O)	I: 0; O: 1	0	0
		UCAS0RXD/UCAS0SOMI	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1

Port P1Sel0.x / P1Sel1.x registers
Select Mode for each Pin in Port

Configuring LED

- To configure LED on P1.0

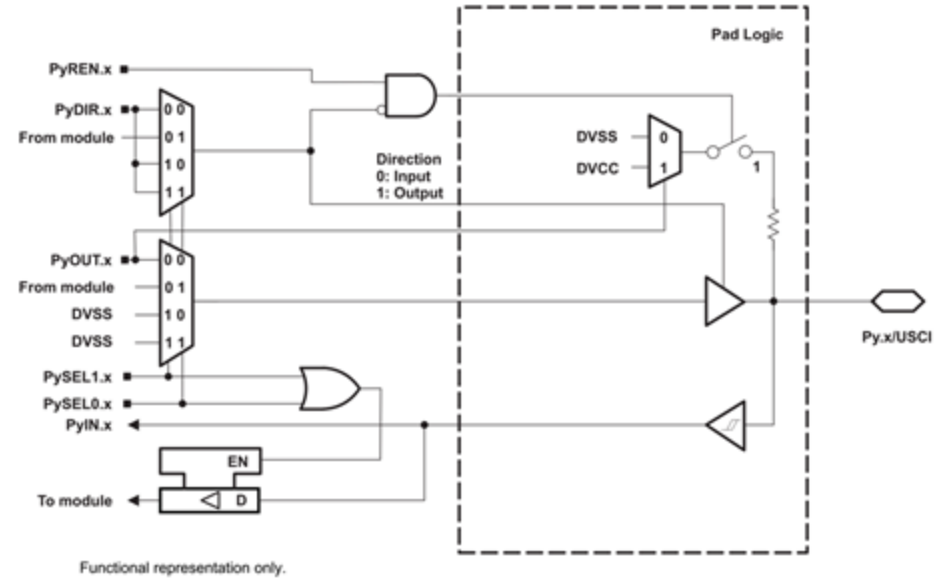
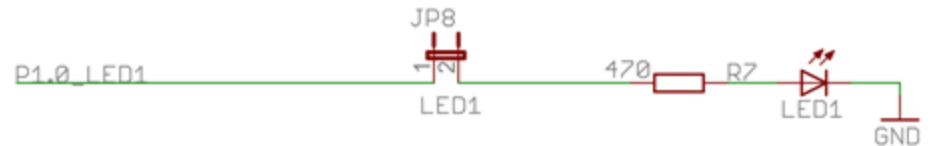


Figure 6-17. Py.x/USCI Pin Schematic



Configuring LED

- To configure LED on P1.0
 - Set direction to output
 - *P1DIR*

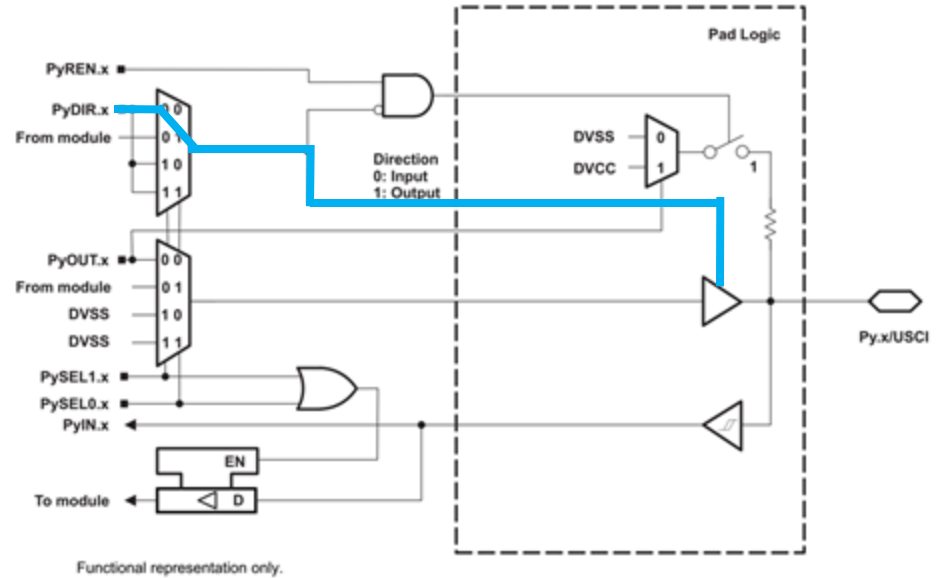
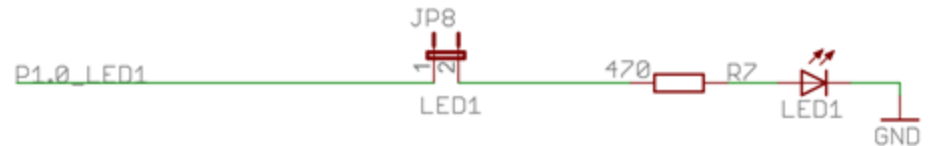


Figure 6-17. Py.x/USCI Pin Schematic



Configuring LED

- To configure LED on P1.0
 - Set direction to output
 - *P1DIR*
 - Set P1.0 output to high
 - *P1OUT*

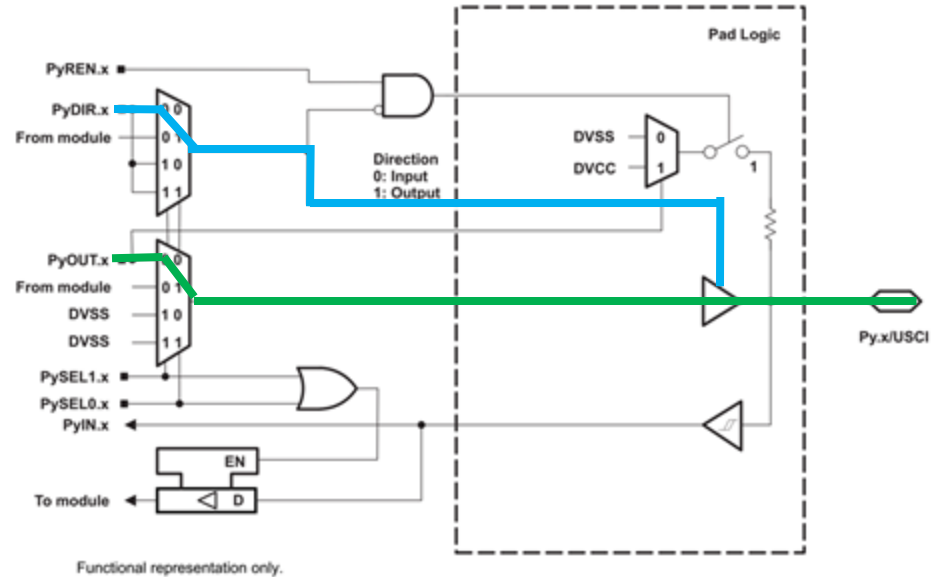
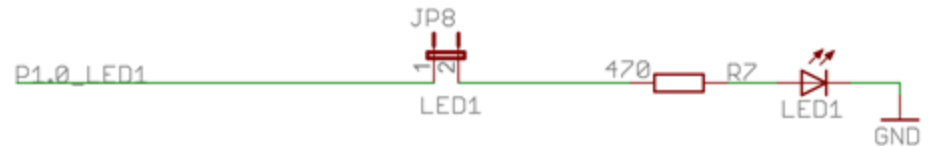


Figure 6-17. Py.x/USCI Pin Schematic



Configuring LED

- To configure LED on P1.0
 - Set direction to output
 - *P1DIR*
 - Set P1.0 output to high
 - *P1OUT*
- To turn on LED, P1.0_LED1
 - voltage needs to be a Logical HIGH
-> VCC(3.3v)

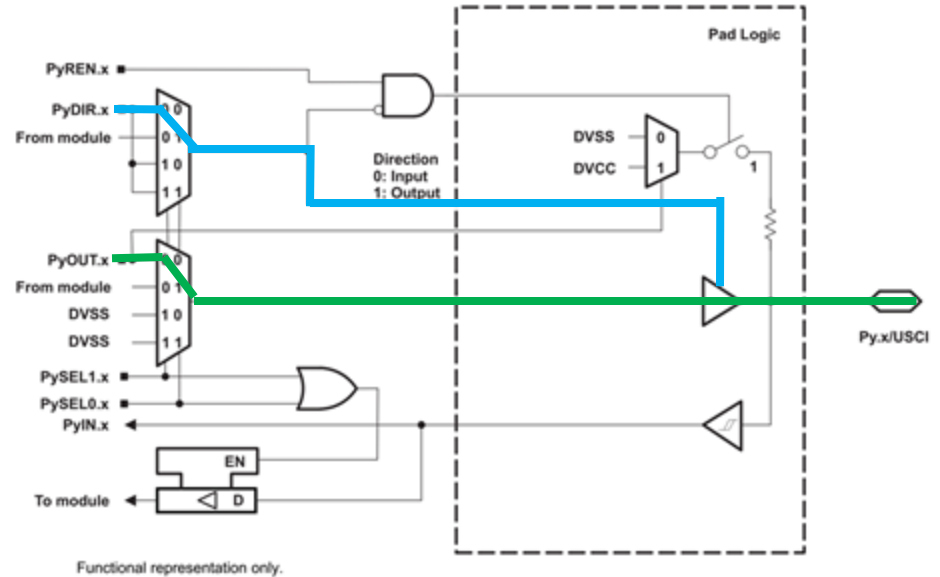
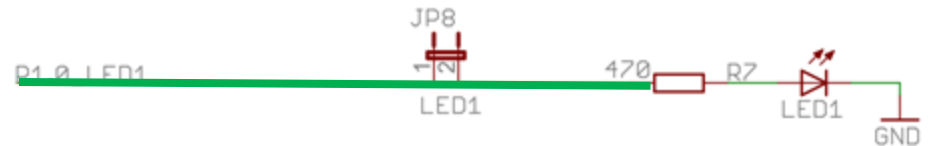


Figure 6-17. Py.x/USCI Pin Schematic



Configuring LED

Also Required to Set Pin to **I/O Mode** (**P1SELx**). But, this is set by default

- To configure LED on P1.0
 - Set direction to output
 - *P1DIR*
 - Set P1.0 output to high
 - *P1OUT*
- To turn on LED, P1.0_LED1
 - voltage needs to be a Logical HIGH
-> VCC(3.3v)

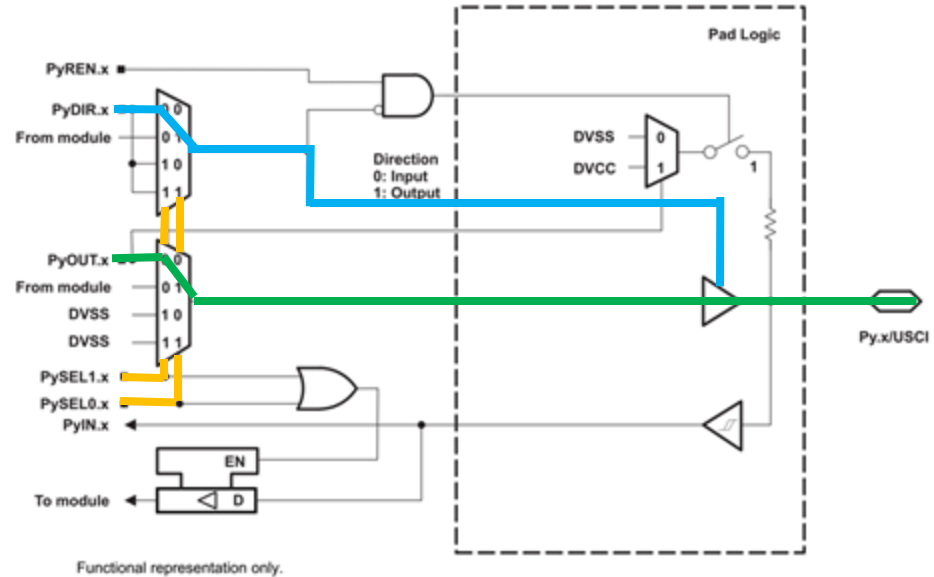
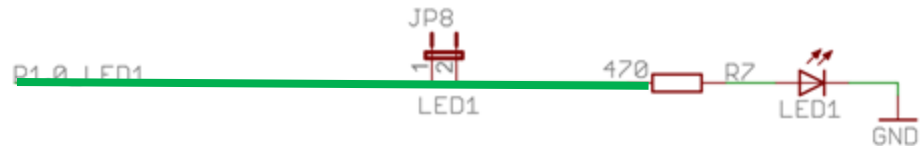


Figure 6-17. Py.x/USCI Pin Schematic



Configuring LED

- Two important registers
 - IO Direction Register (P1DIR)
 - IO Output Register (P1OUT)

Table 12-3. Digital I/O Registers

Offset	Acronym	Register Name	Section
0Eh	P1IV	Port 1 Interrupt Vector	Section 12.4.1
1Eh	P2IV	Port 2 Interrupt Vector	Section 12.4.1
2Eh	P3IV	Port 3 Interrupt Vector	Section 12.4.1
3Eh	P4IV	Port 4 Interrupt Vector	Section 12.4.1
4Eh	P5IV	Port 5 Interrupt Vector	Section 12.4.1
5Eh	P6IV	Port 6 Interrupt Vector	Section 12.4.1
6Eh	P7IV	Port 7 Interrupt Vector	Section 12.4.1
7Eh	P8IV	Port 8 Interrupt Vector	Section 12.4.1
8Eh	P9IV	Port 9 Interrupt Vector	Section 12.4.1
9Eh	P10IV	Port 10 Interrupt Vector	Section 12.4.1
00h	P1IN or PAIN_L	Port 1 Input	Section 12.4.2
02h	P1OUT or PAOUT_L	Port 1 Output	Section 12.4.3
04h	P1DIR or PADIR_L	Port 1 Direction	Section 12.4.4
06h	P1REN or PAREN_L	Port 1 Resistor Enable	Section 12.4.5
08h	P1DS or PADS_L	Port 1 Drive Strength	Section 12.4.6
0Ah	P1SEL0 or PASEL0_L	Port 1 Select 0	Section 12.4.7
0Ch	P1SEL1 or PASEL1_L	Port 1 Select 1	Section 12.4.8
16h	P1SELC or PASELC_L	Port 1 Complement Selection	Section 12.4.9
18h	P1IES or PAIES_L	Port 1 Interrupt Edge Select	Section 12.4.10
1Ah	P1IE or PAIE_L	Port 1 Interrupt Enable	Section 12.4.11
1Ch	P1IFG or PAIFG_L	Port 1 Interrupt Flag	Section 12.4.12

Configuring LED

- Two important registers
 - IO Direction Register (P1DIR)
 - IO Output Register (P1OUT)
- PxOUT, PxDIR
 - x = Port Number
 - Examples: P2OUT, P1DIR

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

7	6	5	4	3	2	1	0
PxOUT							
RW	RW	RW	RW	RW	RW	RW	RW

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

12.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
RW-0	RW-0	RW-0	RW-0	RW-0	RW-0	RW-0	RW-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

P1DIR Register

- Port Direction Register sets the pin to:
 - 0: an input
 - 1: an output

```
/* Configure P1.0 output */  
P1->DIR |= BIT0;
```

rw-0: Bit is Read/Write & defaults to 0

12.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

P1OUT Register

- Port Output Register sets output to:
 - 0: Output set to LOW (GND)
 - 1: Output set to HIGH (VCC)

```
/* Configure P1.0 output */  
P1->DIR |= BIT0;  
  
/* Set P1.0 HIGH */  
P1->OUT |= BIT0;
```

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

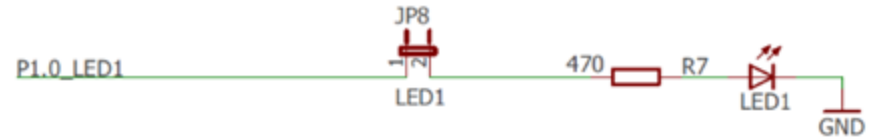
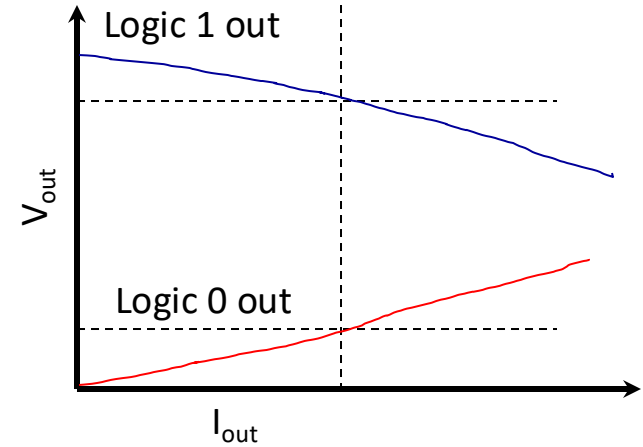
7	6	5	4	3	2	1	0
PxOUT							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

I/O: What's a One? A Zero?

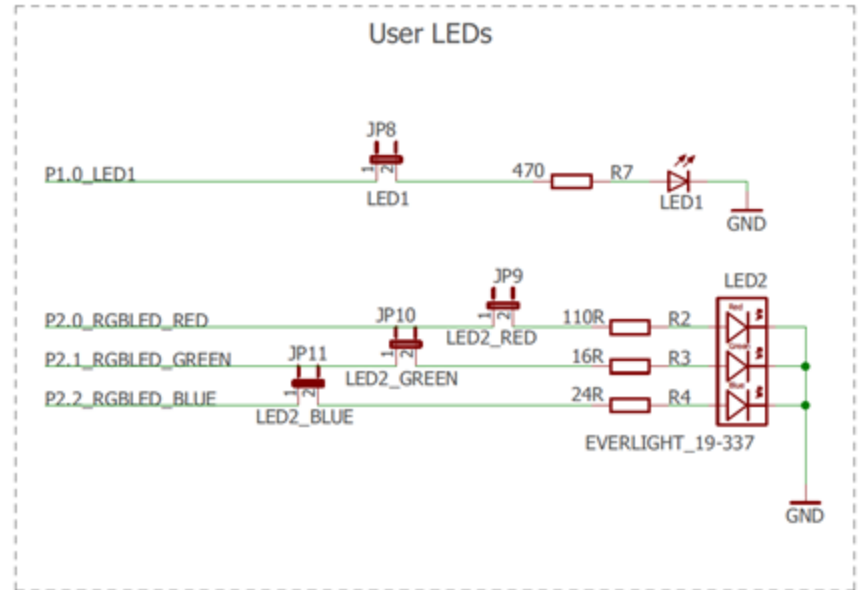
- The input signal's value is determined by the voltage
 - **Exceeding** VDD or GND **may damage the chip**
- Nominal output voltages
 - 1: VDD-0.5V to VDD, 0: 0V to 0.5V
 - **VDD = 3.3V** on MSP432
- E.g., We need to limit current to a value that is safe for both LED and MCU port drive
 - Use **current-limiting resistor**
 - $R = (VDD - V_{LED}) / I_{LED}$
 - $I_{LED} \rightarrow$ Max. LED current
 - Solve for R given $VDD = \sim 3.0V$



Blinking Multiple LEDs

- Which ports?
- Which registers?
 - PxDIR?
 - PxOUT?

Buttons and LEDs



Homework

- Install Software and try **Blink** examples.
- TI CCS: [Download here](#)
 - Download version **12.8.0 (Aug 9, 2024)!**